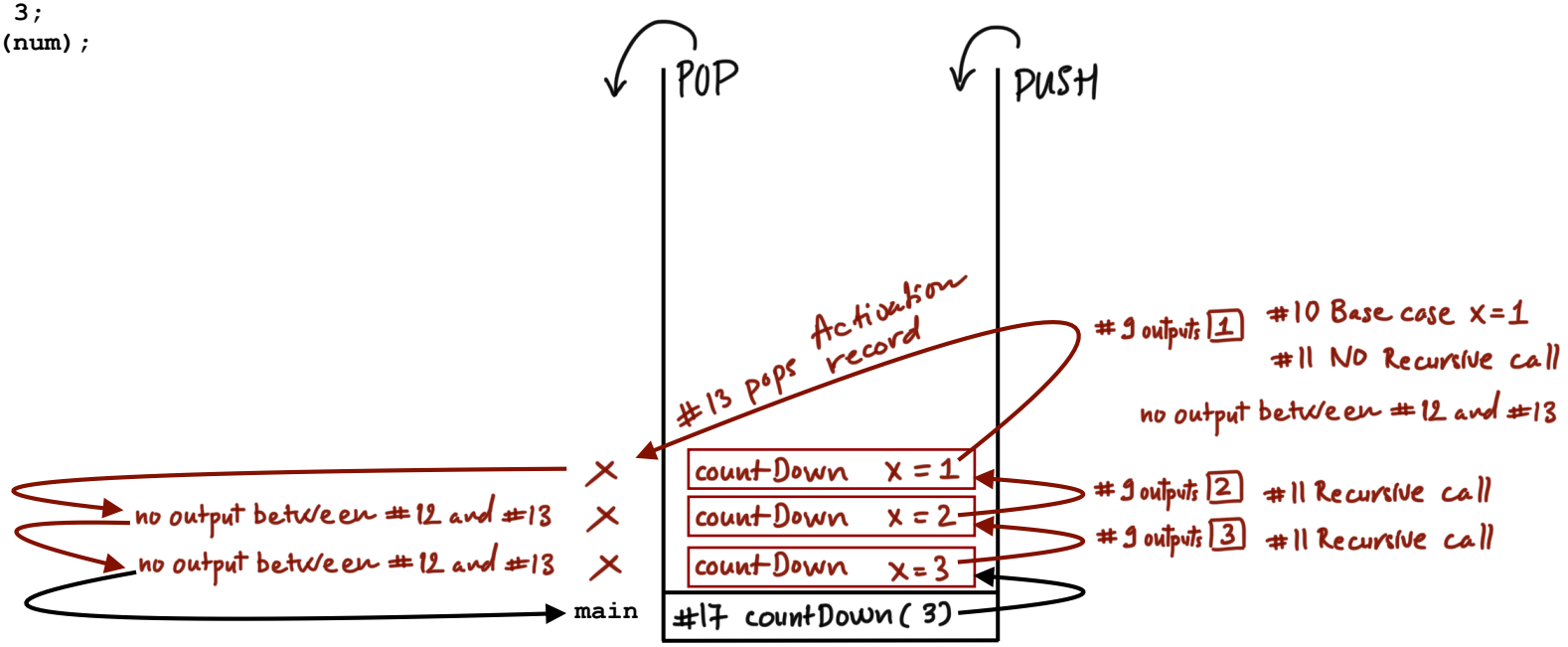
TAIL/DIRECT/IMMEDIATE RECURSION

```
1   package pkg08.practice;
2
3   public class CountDownRecursive {
4         public static void countDown(int x) {
5               System.out.println("x: " + x);
6               if (x > 1) {
7                     countDown(x - 1);
8               }
9         }
10
11        public static void main(String[] args) {
12              int num = 3;
13              countDown(num);
14        }
15  }
16
17  /* OUTPUT
18  x: 3
19  x: 2
20  x: 1
21  */
```



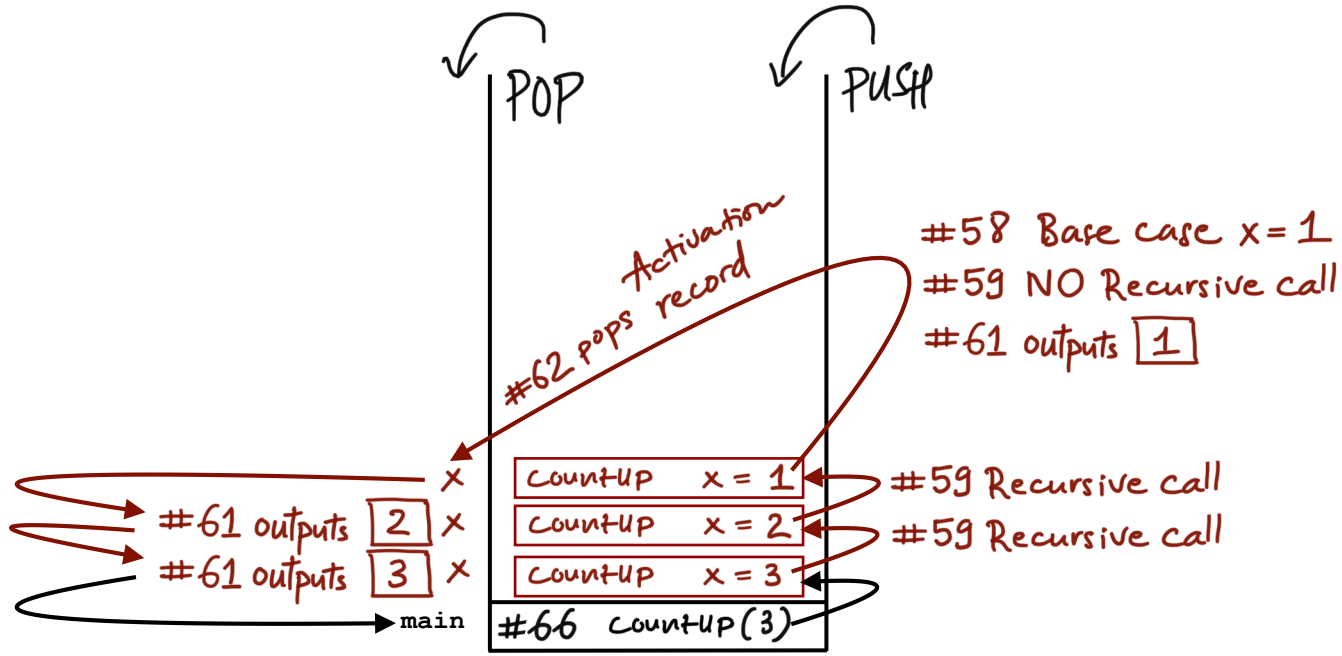
TAIL/DIRECT/IMMEDIATE RECURSION

```
package pkg08.practice;

public class CountUpRecursive {
      public static void countUp(int x) {
            if (x > 1) {
                  countUp(x - 1);
            }
            System.out.println("x: " + x);
      }

      public static void main(String[] args) {
            int num = 3;
            countUp(num);
      }
}

/* OUTPUT
x: 1
x: 2
x: 3
*/
```

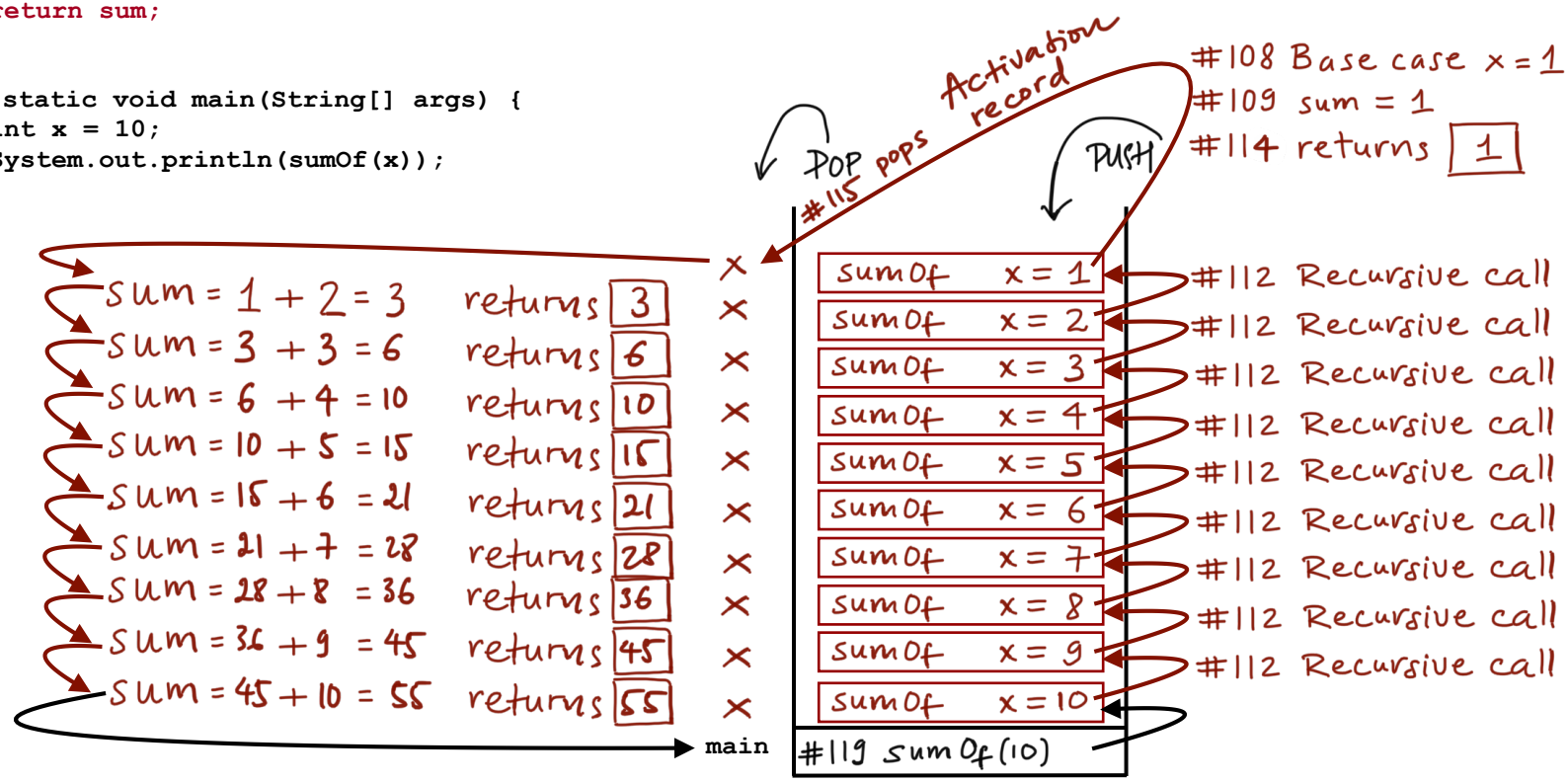


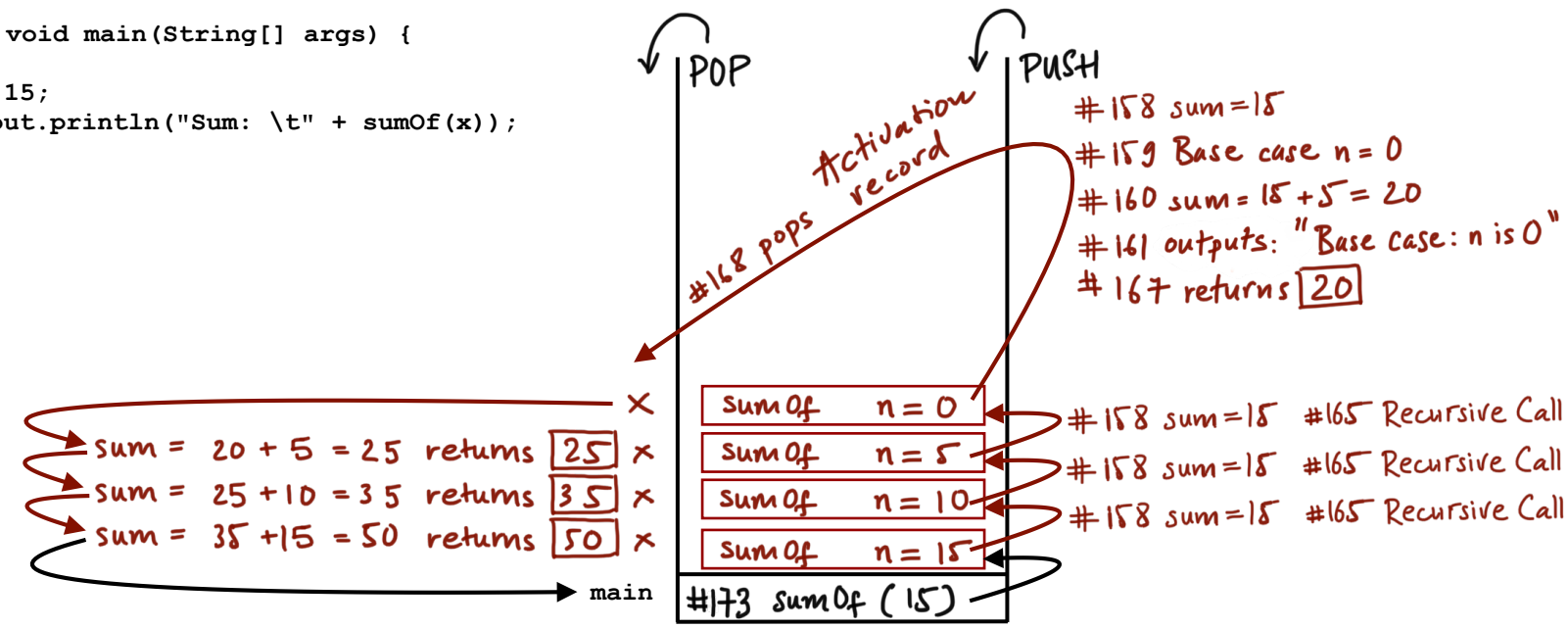
TAIL/DIRECT/IMMEDIATE RECURSION

```
public class SumOf01 {
      public static int sumOf(int x)
            int sum;
            if (x == 1) {
                  sum = 1;
            }
            else {
                  sum = sumOf(x - 1) + x;
            }
            return sum;
      }

      public static void main(String[] args) {
            int x = 10;
            System.out.println(sumOf(x));
      }
}

/* OUTPUT
55
*/
```



TAIL/DIRECT/IMMEDIATE RECURSION

```
package pkg08.practice;

public class SumOf02 {
      public static int sumOf(int n) {
            int sum = 15;
            if (n == 0) {
                  sum += 5;
                  cout << "Base case: n is"
            }
            else
            {
                  sum = sumOf(n - 5) + n;
            }
            return sum;
      }

      public static void main(String[] args) {

            int x = 15;
            System.out.println("Sum: \t" + sumOf(x));
      }
}


/*OUTPUT
Base case: n is 0
Sum:       50
*/
```